



Surgery or Endovascular Treatment in Patients with Anterior Communicating Artery Aneurysm: A Systematic Review and Meta-Analysis

Shahab Aldin Sattari¹, Ataollah Shahbandi², Ryan P. Lee¹, James Feghali¹, Jordina Rincon-Torroella¹, Wuyang Yang¹, Mostafa Abdulrahim¹, Sina Ahmadi¹, Raymond J. So¹, Alice Hung¹, Justin M. Caplan¹, Fernando Gonzalez¹, Rafael J. Tamargo¹, Judy Huang¹, Risheng Xu¹

Key words

- Aneurysm
- Anterior communicating artery
- Clip
- Coil
- Endovascular
- Meta-analysis
- Surgery

Abbreviations and Acronyms

ACoA: Anterior communicating artery

CIN: Certainty in Evidence

mRS: Modified Rankin Scale

OR: Odds ratio

From the ¹Department of Neurosurgery, Johns Hopkins University School of Medicine, Baltimore, Maryland, USA; and ²Tehran School of Medicine, Tehran University of Medical Science, Tehran, Iran

To whom correspondence should be addressed:

Risheng Xu, M.D., Ph.D.

[E-mail: rxu4@jhmi.edu]

Citation: *World Neurosurg.* (2023) 175:31–44.

<https://doi.org/10.1016/j.wneu.2023.03.111>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

1878-8750/\$ - see front matter Published by Elsevier Inc.

INTRODUCTION

The anterior communicating artery (ACoA) is the most common site for both ruptured and unruptured aneurysms, and ACoA aneurysms account for 30%–37% of all intracranial aneurysms.^{1–3} Existing treatment strategies for aneurysms include surgical and endovascular management.³

Surgical approaches were the preferred treatment option for aneurysms until the 1990s,⁴ when the Guglielmi detachable coil were approved for treatment of aneurysms. Since then, endovascular management of aneurysms has gained attention, especially after results of the International Subarachnoid Aneurysm Trial showing higher independent survival when endovascular approaches are employed compared to microsurgical clipping.^{4–8} In contrast, the Barrow

■ **BACKGROUND AND OBJECTIVE:** Although randomized controlled trials have compared surgery versus endovascular treatment for intracranial aneurysms, the literature is sparse in terms of subgroup analysis for anterior communicating artery (ACoA) aneurysm management. This systematic review and meta-analysis sought to compare surgical versus endovascular treatment for ACoA aneurysms.

■ **METHODS:** Medline, PubMed, and Embase were searched from inception to December 12, 2022. Primary outcomes were post-treatment modified Rankin Scale (mRS) >2 and mortality. Secondary outcomes were aneurysm obliteration, retreatment and recurrence, rebleeding, technical failure, vessel rupture, aneurysmal subarachnoid hemorrhage-related hydrocephalus, symptomatic vasospasm, and stroke.

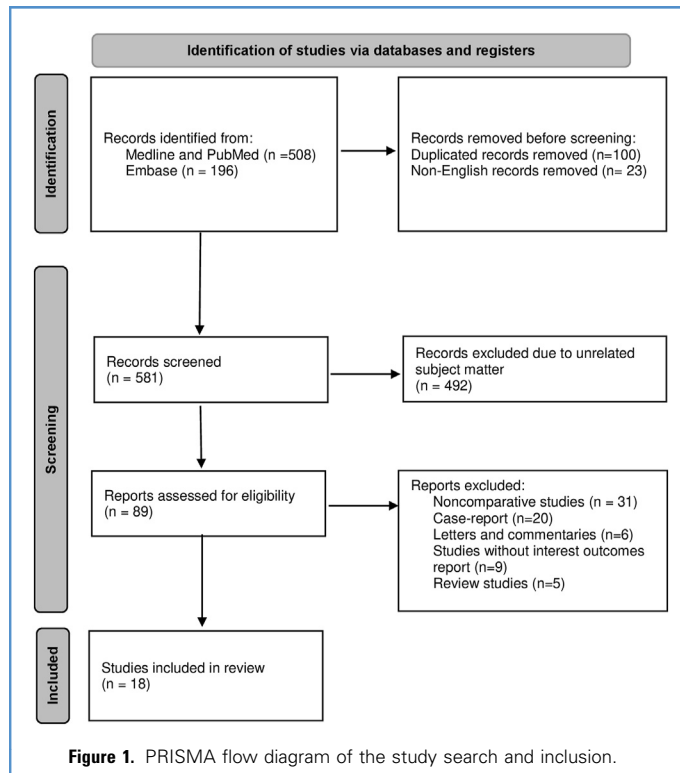
■ **RESULTS:** Eighteen studies yielded 2368 patients, from which 1196 (50.5%) and 1172 (49.4%) patients underwent surgery and endovascular treatment, respectively. The odds ratio (OR) of mortality was similar in total (OR = 0.92 [0.63–1.37], *P* = 0.69), ruptured (OR = 0.92 [0.62–1.36], *P* = 0.66), and unruptured cohorts (OR = 1.58 [0.06–39.60], *P* = 0.78). The OR of mRS > 2 was similar in total (OR = 0.75 [0.50–1.13], *P* = 0.17), ruptured (OR = 0.77 [0.49–1.20], *P* = 0.25), and unruptured cohorts (OR = 0.64 [0.21–1.96], *P* = 0.44). The OR of obliteration was higher with surgery in the total (OR = 2.52 [1.49–4.27], *P* = 0.0008) and ruptured cohorts (OR = 2.61 [1.33–5.10], *P* = 0.005) and unruptured group (OR = 3.46 [1.30–9.20], *P* = 0.01). The OR of retreatment was lower with surgery in the total (OR = 0.37 [0.17–0.76], *P* = 0.007) and ruptured cohorts (OR = 0.31 [0.11–0.89], *P* = 0.03), though it was similar in the unruptured group (OR = 0.51 [0.08–3.03], *P* = 0.46). The OR of recurrence was lower with surgery in the total (OR = 0.22 [0.10, 0.47], *P* = 0.0001), ruptured (OR = 0.16 [0.03, 0.90], *P* = 0.04), and mixed (un) ruptured cohorts (OR = 0.22 [0.09–0.53], *P* = 0.0009). The OR of rebleeding in ruptured group was similar (OR = 0.66 [0.29–1.52], *P* = 0.33). The ORs of other outcomes were similar.

■ **CONCLUSIONS:** ACoA aneurysms may be safely treated with either surgery or endovascular treatment, although microsurgical clipping demonstrates higher obliteration rates and lower rates of retreatment and recurrence.

Ruptured Aneurysm Trial showed no significant difference in postoperative outcomes between endovascular and surgical approaches for aneurysms.^{9–12} As such, the optimal treatment for aneurysms is still unclear,^{3,12–16} although recent evidence suggests that the risk-benefit ratio of both modalities may vary

considerably depending upon the precise location of the aneurysm.^{7–12,17–20}

The literature comparing surgical and endovascular management of ACoA aneurysm in large patient cohorts is sparse.^{21–24} This systematic review and a meta-analysis sought to compare postoperative outcomes in patients with ACoA



aneurysm who underwent either surgical or endovascular treatment.

METHOD

This systematic review was conducted according to the recommendations set forth by the Cochrane Handbook for Systematic Reviews and Interventions²⁵ and followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.²⁶

Eligibility Criteria

Comparative studies with English text that met the following (i.e., patient, intervention, comparator, outcome) were included:

Patient: Individuals with ACoA aneurysm.

Intervention: Surgical treatment.

Comparator: Endovascular treatment.

Outcome: Reporting of either primary outcomes (i.e., post-treatment mortality or modified Rankin Scale²⁷ [mRS] >2) or secondary outcomes (i.e., aneurysm obliteration, retreatment, recurrence, rebleeding, technical failure, vessel

rupture, aneurysmal subarachnoid hemorrhage-related hydrocephalus, symptomatic vasospasm, and stroke).

Search Strategy and Information Source

Two authors (S.A.S. and A.S.) searched Medline, PubMed, and Embase from inception to December 12th, 2022. [("anterior communicating artery" or "AcoA" or "AcomA") and ("aneurysm" or "AcoA aneurysm" or "AcomA aneurysm" or "anterior communication artery aneurysm") and ("surgery" or "surgical" or "clipping" or "surgical clipping" or "surgical clip" or "clip" or "clips") and ("coiling" or "endovascular" or "endovascular coiling" or "coil" or "coils" or "embolization" or "endovascular embolization" or "coil embolization" or "balloon-assisted coiling" or "stent-assisted coiling")]] were used in Medline and PubMed; and [('anterior communicating artery aneurysm'/exp OR 'anterior communicating artery aneurysm') AND ('clipping'/exp OR clipping OR 'microsurgical clipping'/exp OR 'microsurgical clipping') AND ('coiling'/exp OR coiling OR 'endovascular coiling'/exp OR 'endovascular coiling' OR

'endovascular embolization'/exp OR 'endovascular embolization')]] were applied in Embase.

Study Selection

Two authors (S.A.S. and A.S.) screened the records by title and abstract using the eligibility criteria, in which the irrelevant records were excluded, and the remaining studies were reviewed in full text for potential eligibility. The search strategy was supplemented by screening the references of the potentially eligible studies. In case of unresolved discordance, the senior author (R.X.) would adjudicate.

Data Collection Process

Two authors (S.A.S. and A.S.) independently extracted data in 2 preformed Excel sheets, and the files were cross checked and verified against the source material. In case of unresolved discordance, the senior author (R.X.) would adjudicate.

Data Items

The following data were extracted: studies' first author's name, publication year, country, design, number of patients, age, sex, the proportion of unruptured aneurysms, Hunt and Hess grade, Fisher grade, follow-up duration, and outcomes proportion.

Risk of Bias Assessment

Two authors (S.A.S. and A.S.) independently assessed the risk of bias in the included studies using the (i.e., Risk of Bias in Nonrandomized Studies of Interventions) tool.²⁸ In case of any unresolved disagreement in the risk of bias assessment, the senior author (R.X.) would adjudicate. In the case of at least 10 studies per outcome, the publication bias was checked using the Begg's funnel plot.²⁵

Certainty in Evidence

Two authors (S.A.S. and A.S.) independently assessed certainty in evidence (CIN) using the GRADE approach,²⁹ and presented through the GRADEpro GDT Software, 2021, McMaster University and Evidence Prime Inc, Ontario, Canada. In case of any disagreement, the senior author (R.X.) would adjudicate.

Table 1. Baseline Characteristics of the Included Studies

Study and Year	Study Groups	No. of Patients	Age (Year)	Male Sex No. (%)	Unruptured ACoAA No. (%)	Hunt and Hess Grade No. (%)		Fisher Grade No. (%)					Follow-Up (Month)
						I-III	IV-V	0	I	II	III	IV	
Proust et al., 2003 ³⁰	Total	223	49.1*	124 (58)	0 (0)	157 (70)	57 (30)	0 (0)	9 (4)	52 (23)	83 (37)	70 (31)	6–12‡
	Surgery	186	47.3*	106 (60)		133 (71)	45 (29)	0 (0)	9 (5)	44 (23)	64 (34)	61 (32)	
	Endovascular	37	56.7*	18 (51)		24 (72)	12 (28)	0 (0)	0 (0)	8 (21)	19 (51)	9 (24)	
Proust et al., 2009 ³¹	Total	50	46.7*	25 (50)	0 (0)	§	§	§	§	§	§	§	14‡
	Surgery	36	45.4*	17 (47)									
	Endovascular	14	50.1*	8 (57)									
Ahmed et al., 2013 ³²	Total	30	46.3*	14 (46)	0 (0)	30 (100)	0 (0)	NR.	NR.	NR.	NR.	NR.	6–48‡
	Surgery	15				15 (100)	0 (0)						
	Endovascular	15				15 (100)	0 (0)						
Moon et al., 2015 ³³	Total	130	52.5*	78 (60)	0 (0)	NR.	NR.	NR.	NR.	NR.	NR.	NR.	12–36‡
	Surgery	91											
	Endovascular	39											
Heit et al., 2017 ³⁴	Total	100	52.5*	43 (43)	0 (0)	83 (83)	17 (17)	1 (1)	2 (2)	10 (10)	48 (48)	39 (30)	3*
	Surgery	50	50*	24 (48)		47 (94)	3 (6)	1 (2)	2 (4)	6 (12)	29 (58)	12 (24)	
	Endovascular	50	55*	19 (38)		36 (72)	14 (28)	0 (0)	0 (0)	4 (8)	19 (38)	27 (54)	
Kumar et al., 2017 ³⁵	Total	50	50.9*	28 (56)	0 (0)	50 (100)	0 (0)	NR.	NR.	NR.	NR.	NR.	6*
	Surgery	33	49.7*	17 (51)		33 (66)	0 (0)						
	Endovascular	17	52.1*	11 (64)		17 (34)	0 (0)						
Steklacova et al., 2017 ³⁶	Total	398	55*	191 (47)	94 (23)	206 (51)	98 (24)	0 (0)	17 (4)	69 (17)	119 (29)	79 (19)	6–12‡
	Surgery	79	51*		25 (31)	43 (54)	11 (14)	0 (0)	2 (2)	21 (26)	21 (26)	10 (12)	
	Endovascular	319	56*		69 (21)	163 (51)	87 (27)	0 (0)	15 (4)	68 (21)	98 (30)	69 (21)	

ACoAA, anterior communicating artery aneurysm; No, number; NA, not applicable; NR, not reported.

*Mean.

‡Median.

‡Range.

§In Proust et al. 2009 study, the surgery group contained 23 (69%) Hunt and Hess (HH) grade I-II, 13 (36%) HH grade III-IV, 12 (33%) Fisher grade I-II, 13 (36%) Fisher grade III-IV, and the endovascular group included 8 (57) HH grade I-II, 6 (42%) HH grade III-IV, 9 (64%) Fisher grade I-II, 5 (35%) Fisher grade III-IV. In Zhao et al. 2019 study, the surgery group included 13 (20%) Fisher grade I-II, 52 (80%) Fisher grade IV-V, and the endovascular group contained 11 (24%) Fisher grade I-II and 35 (76%) Fisher grade IV-V. Backer et al., 2020 and Baykara et al., 2021 studies had a short-term in-hospital follow-up.

||Included studies were retrospective observational studies except Proust et al., 2003,³⁰ Proust et al., 2009³¹ and Ahmed et al., 2013,³² which were prospective.

Continues

Table 1. Continued

Study and Year	Study Groups	No. of Patients	Age (Year)	Male Sex No. (%)	Unruptured ACoAA No. (%)	Hunt and Hess Grade No. (%)		Fisher Grade No. (%)					Follow-Up (Month)
						I-III	IV-V	0	I	II	III	IV	
Nassiri et al., 2018 ³⁷	Total	36	53.9*	18 (50)	0 (0)	30 (83)	6 (17)	0 (0)	7 (19)	7 (19)	8 (22)	14 (38)	96*
	Surgery	27	56.9*	13 (48)		21 (77)	6 (23)	0 (0)	5 (18)	5 (18)	8 (29)	9 (33)	
	Endovascular	9	50.9*	5 (55)		9 (100)	0 (0)	0 (0)	2 (22)	2 (22)	0 (0)	5 (55)	
Schmalz et al., 2018 ³⁸	Total	149	61*	54 (36)	149 (100)	NA.	NA.	NA.	NA.	NA.	NA.	NA.	12*
	Surgery	98											
	Endovascular	51											
Zhao et al., 2019 ³⁹	Total	111	55.1*	51 (45)	0 (0)	NR.	NR.	§	§	§	§	§	32*
	Surgery	65	55.5*	29 (44)									
	Endovascular	46	45.5*	22 (47)									
Backer et al., 2020 ⁴⁰	Total	116	50†	57 (49)	0 (0)	112 (96)	14 (4)	NR.	NR.	NR.	NR.	NR.	§
	Surgery	45	53†	24 (53)		42 (93)	3 (7)						
	Endovascular	71	49†	33 (46)		60 (84)	11 (16)						
Ki et al., 2020 ⁴¹	Total	214	56.8*	110 (51)	116 (54)	NR.	NR.	NR.	NR.	NR.	NR.	NR.	36*
	Surgery	105											
	Endovascular	109											
Moon et al., 2020 ⁴²	Total	70	61.2*	39 (55)	70 (100)	NA.	NA.	NA.	NA.	NA.	NA.	NA.	6–12‡
	Surgery	33	64.8*	23 (69)									
	Endovascular	37	57.6*	16 (43)									
Baykara et al., 2021 ⁴³	Total	92	53.9*	58 (63)	0 (0)	NR.	NR.	0 (0)	0 (0)	0 (0)	92 (100)	0 (0)	§
	Surgery	40	51*	23 (57.5)				0 (0)	0 (0)	0 (0)	40 (100)	0 (0)	
	Endovascular	52	56.9*	35 (67.3)				0 (0)	0 (0)	0 (0)	52 (100)	0 (0)	
Harris et al., 2021 ⁴⁴	Total	132	57.7*	53 (40)	0 (0)	NR.	NR.	0 (0)	6 (4)	14 (10)	18 (13)	64 (48)	36†
	Surgery	19	57.9*	7 (36)				0 (0)	0 (0)	1 (5)	4 (21)	11 (57)	
	Endovascular	113	57.6*	46 (40)				0 (0)	6 (5)	13 (11)	14 (12)	53 (47)	
Ma et al., 2021 ⁴⁵	Total	126	49.2*	76 (60)	0 (0)	126 (100)	0 (0)	NR.	NR.	NR.	NR.	NR.	38*
	Surgery	99		63 (63)		99 (100)	0 (0)						
	Endovascular	27		13 (48)		27 (100)	0 (0)						

Haeren et al., 2022 ⁴⁶	Total	128	60†	67 (52)	128 (100)	NA.	NA.	NA.	NA.	NA.	NA.	NA.	6*
	Surgery	81	59†	41 (51)									
	Endovascular	47	61†	26 (55)									
Lee et al., 2022 ⁴⁷	Total	213	57.3*	112 (52.6)	0 (0)	NR.	NR.	0 (0)	55 (25.8)	30 (14.1)	61 (28.6)	67 (31.5)	3*
	Surgery	94	53.5*	46 (49)				0 (0)	18 (19.1)	18 (19.1)	31 (33)	27 (28.8)	
	Endovascular	119	60.3*	66 (55.5)				0 (0)	37 (31.1)	12 (10.1)	30 (25.2)	40 (33.6)	

ACoAA, anterior communicating artery aneurysm; No, number; NA, not applicable; NR, not reported.

*Mean.

†Median.

‡Range.

§In Proust et al. 2009 study, the surgery group contained 23 (69%) Hunt and Hess (HH) grade I-II, 13 (36%) HH grade III-IV, 12 (33%) Fisher grade I-II, 13 (36%) Fisher grade III-IV, and the endovascular group included 8 (57) HH grade I-II, 6 (42%) HH grade III-IV, 9 (64%) Fisher grade I-II, 5 (35%) Fisher grade III-IV. In Zhao et al. 2019 study, the surgery group included 13 (20%) Fisher grade I-II, 52 (80%) Fisher grade IV-V, and the endovascular group contained 11 (24%) Fisher grade I-II and 35 (76%) Fisher grade IV-V. Backer et al., 2020 and Baykara et al., 2021 studies had a short-term in-hospital follow-up.

||Included studies were retrospective observational studies except Proust et al., 2003,³⁰ Proust et al., 2009³¹ and Ahmed et al., 2013,³² which were prospective.

Table 2. Risk of Bias Assessment

Study and Year	Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) Domains						
	1	2	3	4	5	6	7
Proust et al., 2003 ³²							
Proust et al., 2009 ³³							
Ahmed et al., 2013 ³⁴							
Moon et al., 2015 ³⁵							
Heit et al., 2017 ³⁶							
Kumar et al., 2017 ³⁷							
Steklacova et al., 2017 ³⁸							
Nassiri et al., 2018 ³⁹							
Schmalz et al., 2018 ⁴⁰							
Zhao et al., 2019 ⁴¹							
Backer et al., 2020 ⁴²							
Ki et al., 2020 ⁴³							
Moon et al., 2020 ⁴⁴							
Harris et al., 2021 ⁴⁵							
Ma et al., 2021 ⁴⁶							
Haeren et al., 2022 ⁴⁶							
Lee et al., 2022 ⁴⁷							

ROBINS-I Domains: 1, Baseline confounders; 2, Participants' enrollment; 3, Interventions' classification; 4, Deviation from intended interventions; 5, Incomplete outcome data; 6, Measurement of the outcomes; 7, selective reporting. Interpretation of colors: Black, low risk; Light gray, moderate risk; Dark gray, high risk.

Statistical Analysis

Baseline characteristics were presented using descriptive statistics. The pooled odds ratio (OR) of categorical data with a corresponding 95% confidence interval (CI) was calculated through the Mantel-Haenszel method using the Review Manager, version 5.4.1. The heterogeneities between studies were detected using the I^2 test statistic, which attributes the proportion of total variation to the differences between studies rather than sampling error.²⁵ In cases of substantial heterogeneity (i.e., $I^2 > 50\%$ and $P < 0.10$), the random-effect model was used; otherwise, the fixed-effect model was applied. Given the included studies represented ruptured, unruptured, and the mixed cohort of aneurysms, in the case of unruptured aneurysms per

outcome, subgroup analysis was performed in which the exclusively ruptured cohorts were analyzed separately from the mixed cohorts. All statistical tests were conducted with a two-tailed statistical significance set at a $P < 0.05$.

RESULTS

Study Selection

The applied search strategy resulted in 704 records, from which 23 non-English records and 100 duplicate records were removed. Five hundred eighty-one records were screened by title and abstract and identified 89 relevant studies. Of these, 71 articles were excluded as they were non-comparative studies ($n = 31$), case reports ($n = 20$), letters and commentaries ($n =$

6), studies without outcomes of interest ($n = 9$), or review studies ($n = 5$). Therefore, 18 studies were ultimately included in this systematic review and meta-analysis (Figure 1).³⁰⁻⁴⁷

Study Characteristics

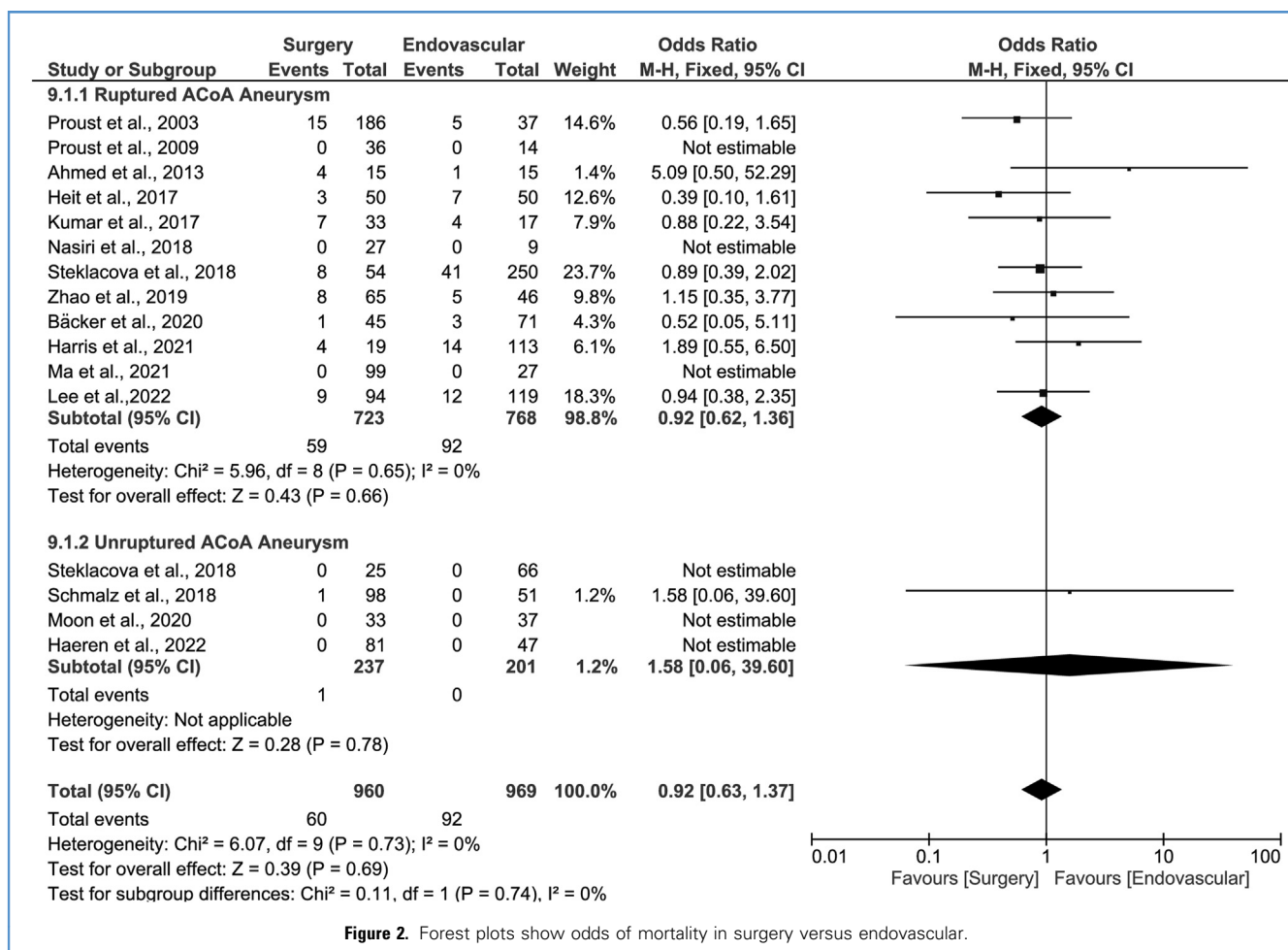
The 18 studies were published between 2003 and 2022 and were conducted in the United States ($n = 3$), Canada ($n = 1$), United Kingdom ($n = 1$), France ($n = 2$), Czech Republic ($n = 1$), Hungary ($n = 1$), Finland ($n = 1$), Egypt ($n = 1$), Turkey ($n = 1$), India ($n = 1$), China ($n = 2$), and Korea ($n = 3$). In total, the studies yielded 2368 patients, of which 1196 (50.5%) and 1172 (49.4%) patients underwent surgery and endovascular treatment, respectively. Across the 18 studies, the mean age of patients ranged from 45.4 to 64.8 years, and 1198 (50.5%) patients were male. Additionally, 557 (23.5%) patients presented with unruptured aneurysms, and the median (range) follow-up duration was 9 (3–96) months (Table 1).

Risk of Bias Assessment

In terms of baseline confounders, Steklacova et al.,³⁶ Schmalz et al.,³⁸ Backer et al.,⁴⁰ and Moon et al.⁴² were found to be at high risk of bias; the remaining studies were at moderate risk. In terms of the remaining domains of the Risk of Bias In Non-randomized Studies of Interventions, the included studies were at moderate risk of bias (Table 2).

Results of Meta-Analysis

Primary Outcomes. Mortality. Mortality was reported in 15 studies. In the surgery group ($n = 960$), 60 instances of mortality were reported, whereas 92 instances occurred in endovascular ($n = 969$) group. The results of this meta-analysis showed that the odds of mortality were not significantly different between the 2 groups (6.2% vs. 9.4%, OR = 0.92 [0.63–1.37], $P = 0.69$, $I^2 = 0\%$, $P = 0.73$, CIN: moderate; Figure 2). Publication bias was checked using the funnel plot, which was symmetric and thereby indicated no publication bias (Figure 3). In the subgroup of patients with ruptured aneurysms, mortality was reported in 59 patients for the surgery cohort ($n = 723$) and 92 patients for the endovascular



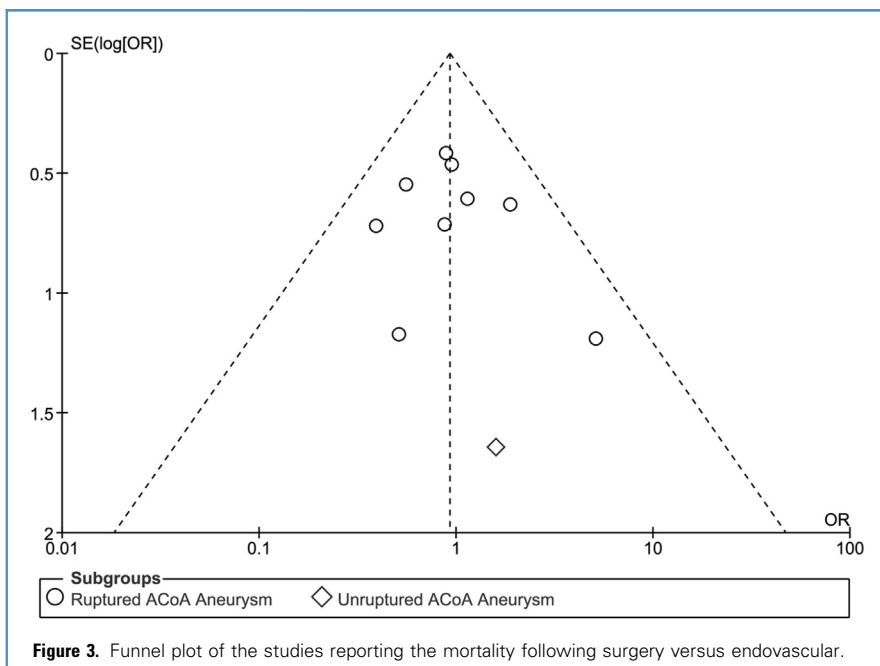
group (n = 768). The meta-analysis showed no significant difference in the odds of mortality between these 2 groups (8.1% vs. 11.9%, OR = 0.92 [0.62–1.36], P = 0.66, I² = 0%, P = 0.65, CIN: moderate; **Figure 2**). In the subgroup of patients with unruptured aneurysms, mortality was reported in 1 patient for the surgery cohort (n = 237) and 0 patients for the endovascular group (n = 201). The meta-analysis showed no significant difference in the odds of mortality between these 2 groups (0.4% vs. 0%, OR = 1.58 [0.06–39.6], P = 0.78, CIN: low; **Figure 2**).

Follow-Up mRS > 2. Eight studies reported follow-up mRS, and the number of cases with mRS > 2 were tallied. In total, 88 instances with mRS > 2 were reported in the surgical cohort (n = 453), whereas 92

cases were reported for the endovascular group (n = 305) groups. The results of this meta-analysis showed that the odds of mRS > 2 were not significantly different between the 2 approaches (19.4% vs. 30.1%, OR = 0.75 [0.50–1.13], P = 0.17, I² = 0%, P = 0.72, CIN: moderate; **Figure 4**). In the subgroup of patients with ruptured aneurysms, 81 cases of mRS > 2 were noted for the surgery group (n = 322) and 86 cases for the endovascular cohort (n = 217) groups. The odds of follow-up mRS > 2 were similar between the 2 groups using this meta-analysis (25.1% vs. 39.6%, OR = 0.77 [0.49–1.20], P = 0.25, I² = 0%, P = 0.51, CIN: moderate; **Figure 4**). In the subgroup of patients with unruptured aneurysms, 7 cases of mRS > 2 were noted for the surgery group (n = 131), and 6 cases for the endovascular

cohort (n = 88) groups. The odds of follow-up mRS > 2 were similar between the 2 groups using this meta-analysis (5.3% vs. 6.8%, OR = 0.64 [0.21–1.96], P = 0.44, I² = 0%, P = 0.70, CIN: low; **Figure 4**).

Secondary Outcomes. Aneurysm Obliteration. Seven studies reported aneurysm obliteration. In the surgical cohort (n = 438), 400 cases of aneurysm obliteration were observed, compared to 510 instances in the endovascular group (n = 573) group. This meta-analysis showed that the odds of obliteration were significantly higher following surgical management (91.3% vs. 89% OR = 2.52 [1.49–4.27], P = 0.0006, I² = 0%, P = 0.54, CIN: high; **Figure 5**). In the subgroup of patients with ruptured aneurysms, 239 and 136 cases of

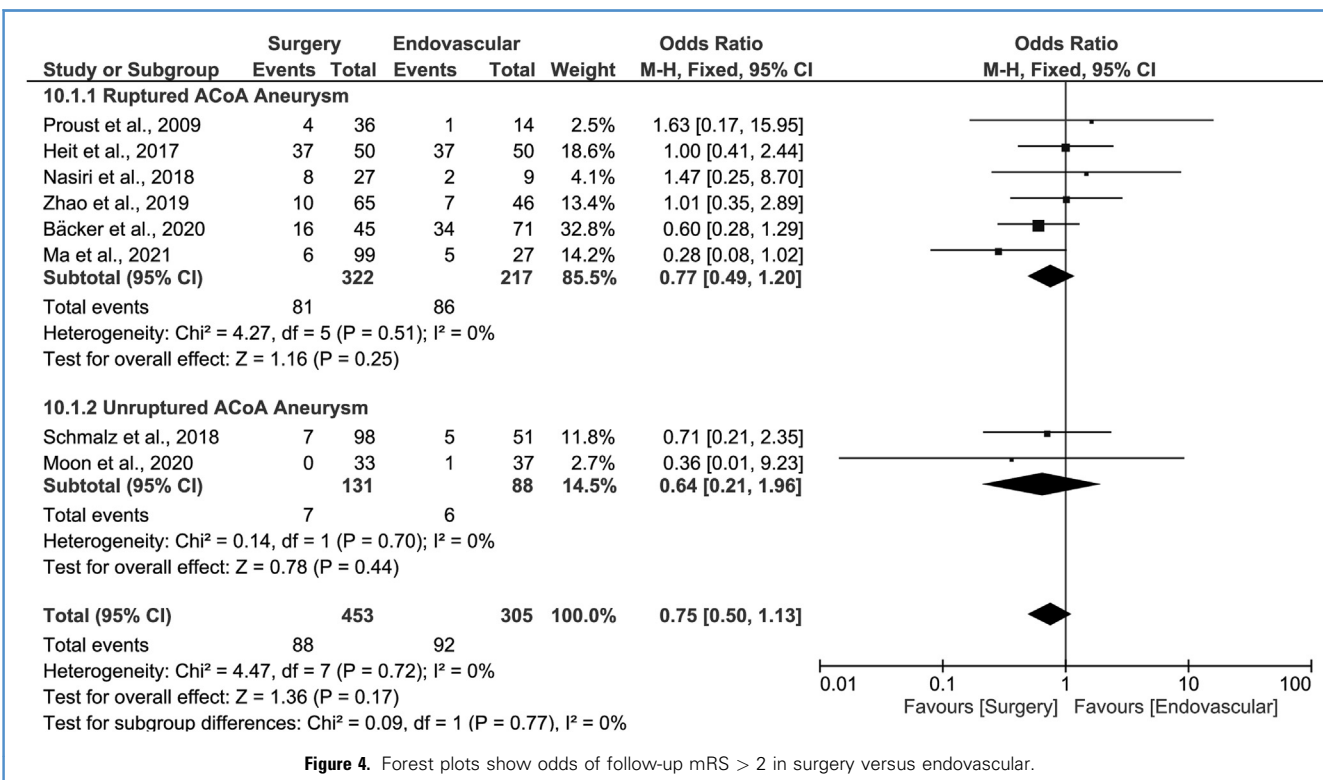


aneurysm obliteration were noted in the surgery (n = 268) and endovascular (n = 173) groups, respectively. The meta-analysis on this subgroup similarly

showed higher odds of aneurysm obliteration for the surgery group (89.1% vs. 78.6%, OR = 2.61 [1.33–5.10], P = 0.005, I² = 0%, P = 0.43, CIN: high; Figure 5). In

the subgroup of patients with unruptured aneurysms, 84 and 65 cases of aneurysm obliteration were noted in the surgery (n = 91) and endovascular (n = 84) groups, respectively. The meta-analysis on this subgroup similarly showed higher odds of aneurysm obliteration for the surgery group (92.3% vs. 77.3%, OR = 3.46 [1.30–9.20], P = 0.01, I² = 0%, P = 0.73, CIN: moderate; Figure 5).

Re-Treatment. Re-treatment was reported in 7 studies. In the surgery group (n = 395), 14 cases of re-treatment were observed, compared to 54 cases for the endovascular group (n = 607). Using this meta-analysis, the odds of re-treatment was found to be significantly lower for the surgery cohort (3.5% vs. 8.8%, OR = 0.37 [0.17–0.76], P = 0.007, I² = 0%, P = 0.96, CIN: moderate; Figure 6). For the subgroup of patients presenting with ruptured aneurysms, 9 and 22 cases of re-treatment were noted in the surgery (n = 202) and endovascular (n = 207) groups, respectively. As before, the odds of re-treatment remained significantly lower for patients who underwent surgery (4.4% vs. 10.6%, OR = 0.31 [0.11–0.89], P = 0.03, I² = 0%, P = 0.84, CIN: moderate; Figure 6). For the subgroup of patients



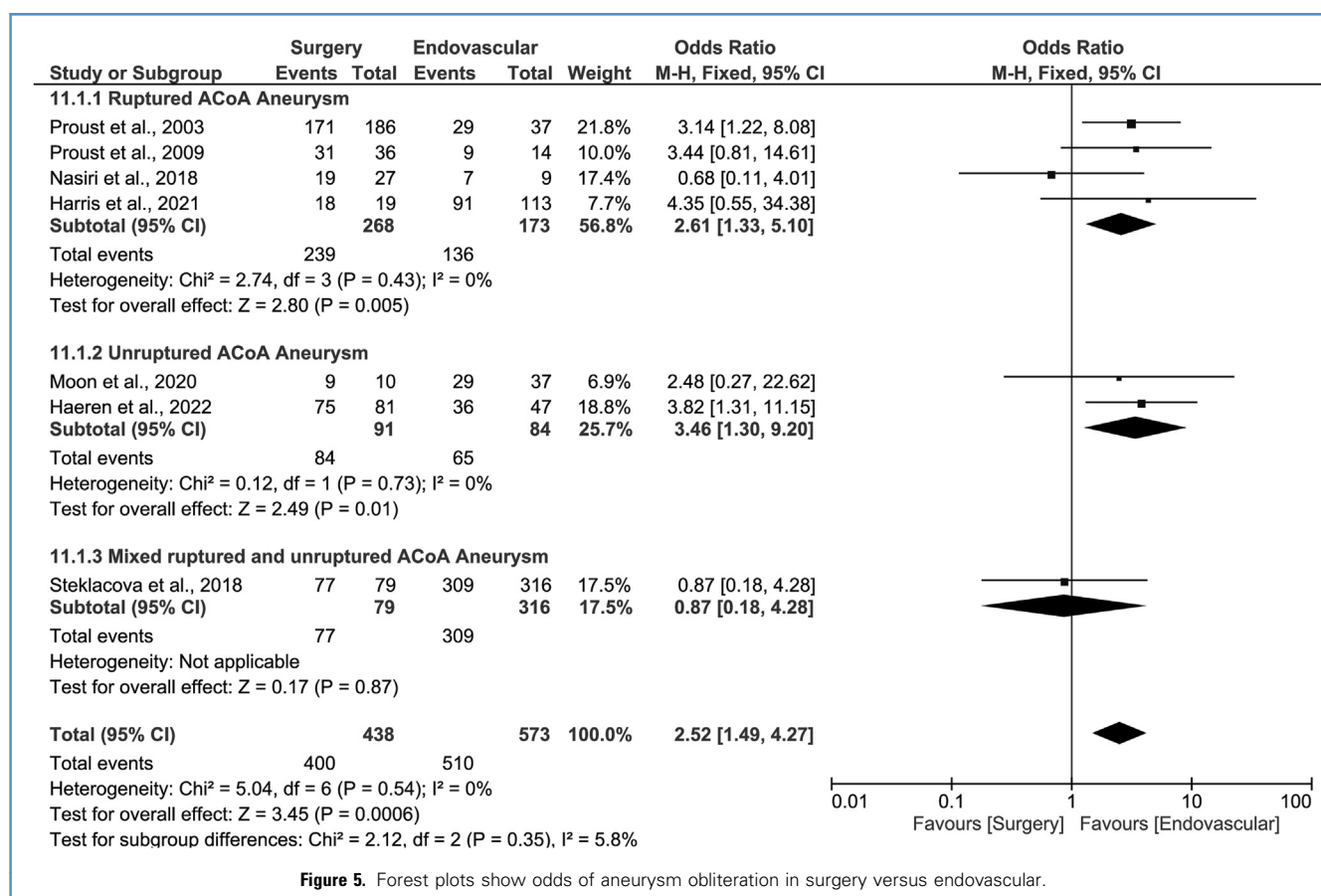


Figure 5. Forest plots show odds of aneurysm obliteration in surgery versus endovascular.

presenting with unruptured aneurysms, 2 and 3 cases of re-treatment were noted in the surgery ($n = 114$) and endovascular ($n = 84$) groups, respectively. The odds of retreatment was lower within the surgery group, though not significantly (1.7% vs. 3.5%, OR = 0.51 [0.08–3.03], $P = 0.46$, $I^2 = 0\%$, $P = 0.46$, CIN: low; Figure 6).

Recurrence. Recurrence was reported by 5 studies, and 8 and 64 cases of recurrence were noted in the surgery ($n = 251$) and endovascular ($n = 590$) groups, respectively. The results of this meta-analysis showed that the odds of recurrence was significantly lower following surgery (3.1% vs. 10.8%, OR = 0.22 [0.10–0.47], $P = 0.0001$, $I^2 = 0\%$, $P = 0.76$, CIN: moderate; Figure 7). In the subgroup of patients with ruptured aneurysms, 1 case of recurrence was observed for the surgery group ($n = 34$), whereas 27 cases were noted in the endovascular cohort ($n = 128$). On this metanalysis, the odds of recurrence

remained significantly lower for patients who underwent surgery (OR = 0.16 [0.03–0.90], $P = 0.04$, $I^2 = 0\%$, $P = 0.60$, CIN: moderate; Figure 7).

Rebleeding. Eight studies reported rebleeding after treatment of ruptured ACoA. In the surgery group ($n = 563$), 10 cases of rebleeding were observed, compared to 18 cases in the endovascular group ($n = 448$). Using this meta-analysis, the odds of rebleeding was similar between the groups (OR = 0.66 [0.29–1.52], $P = 0.33$, $I^2 = 0\%$, $P = 0.66$, Figure 8).

Technical Aspects and Periprocedural Morbidities. Two studies reported technical failure, in which the OR was similar in the mixed ruptured and unruptured cohort (4.5% vs. 8.1%, OR = 0.55 [0.18–1.63], $P = 0.28$, $I^2 = 0\%$, $P = 0.65$). Seven studies reported intraoperative vessel rupture, in which the ORs were similar in overall (16.3% vs. 3.9%, OR = 2.54 [0.69–8.69], $P = 0.17$, $I^2 = 69\%$, $P = 0.004$), ruptured (19.7% vs. 4.3%,

OR = 2.37 [0.48–11.75], $P = 0.29$, $I^2 = 79\%$, $P = 0.0009$), and unruptured cohorts (5% vs. 2%, OR = 2.30 [0.34–15.59], $P = 0.39$, $I^2 = 0\%$, $P = 0.55$). Three studies reported aneurysmal subarachnoid hemorrhage-related hydrocephalus after ACoA rupture, in which the OR was similar (9.8% vs. 2.8%, OR = 2.87 [0.69–11.92], $P = 0.15$, $I^2 = 0\%$, $P = 0.84$). Eight studies reported symptomatic vasospasm after ACoA rupture, in which the OR was higher in the surgery group (21.4% vs. 18.2%, OR = 1.47 [1.05–2.06], $P = 0.02$, $I^2 = 23\%$, $P = 0.25$). Four studies on ruptured ACoA reported stroke, in which the OR was similar (7.2% vs. 20.4%, OR = 0.69 [0.15–3.29], $P = 0.64$, $I^2 = 69\%$, $P = 0.29$).

DISCUSSION

Despite recent advances in surgical and endovascular approaches for cerebral aneurysms, the optimal approach for

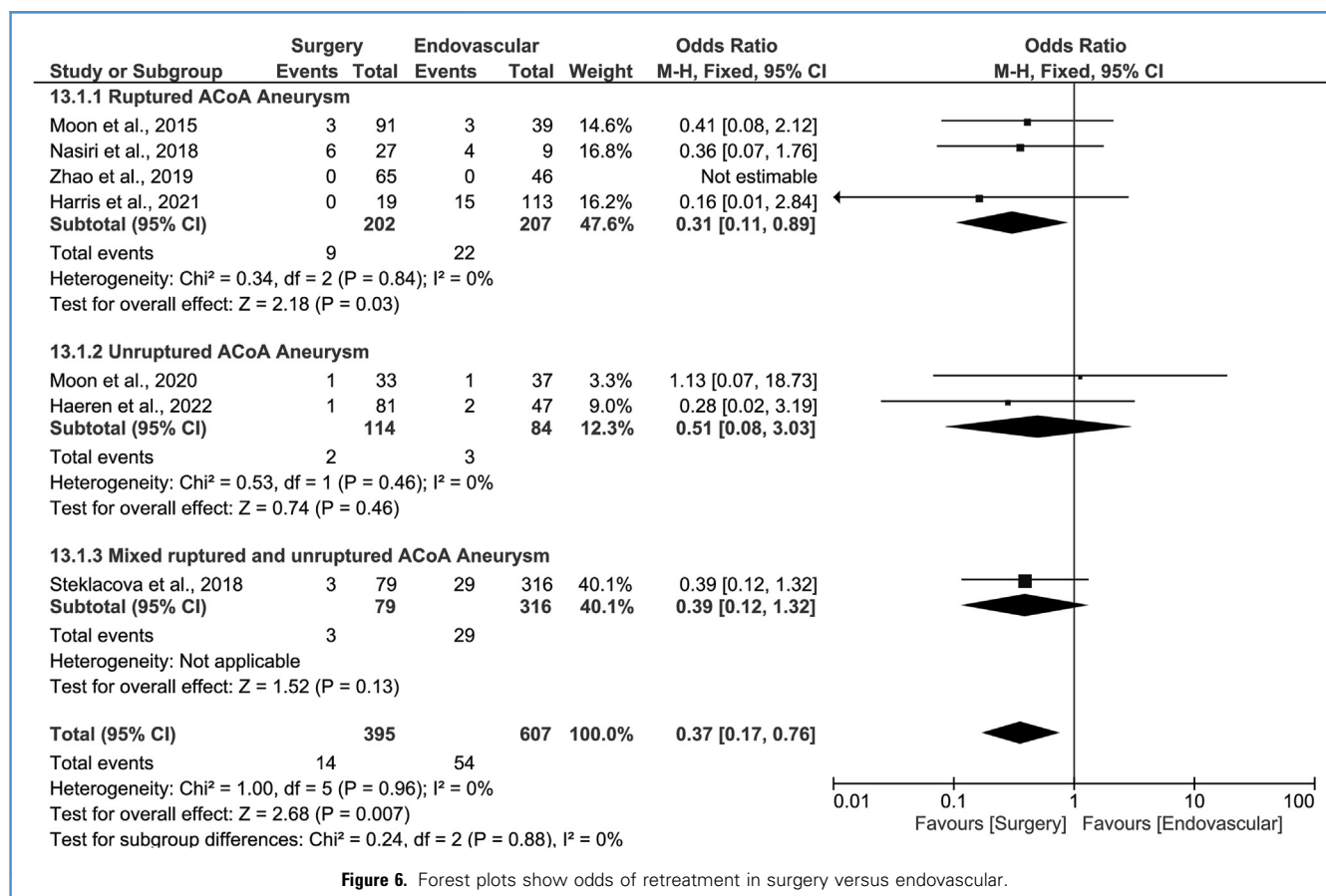


Figure 6. Forest plots show odds of retreatment in surgery versus endovascular.

management of ACoA aneurysm remains unclear.^{4,12-16} Although the ACoA is a frequent location for both ruptured and unruptured aneurysms,³ surgical approaches for an ACoA aneurysm may be challenging due to complex anatomical factors such as dome projection, proximity to critical structures such as perforators and the optic nerve, and the involvement of branch vessels.^{21,22,48} Conversely, an endovascular approach may be less preferable in cases involving hematomas necessitating surgical decompression, small size (<3 mm), giant, or multiple aneurysms, or an unfavorable dome-to-neck ratio or complex arterial tortuosity.^{23,24,30,33}

To date, the literature comparing surgical and endovascular management of ACoA aneurysm is sparse.^{7-12,17} This systematic review and a meta-analysis compared outcomes in patients with ACoA aneurysm who underwent either surgical or endovascular management of their condition.

Summary of Key Findings

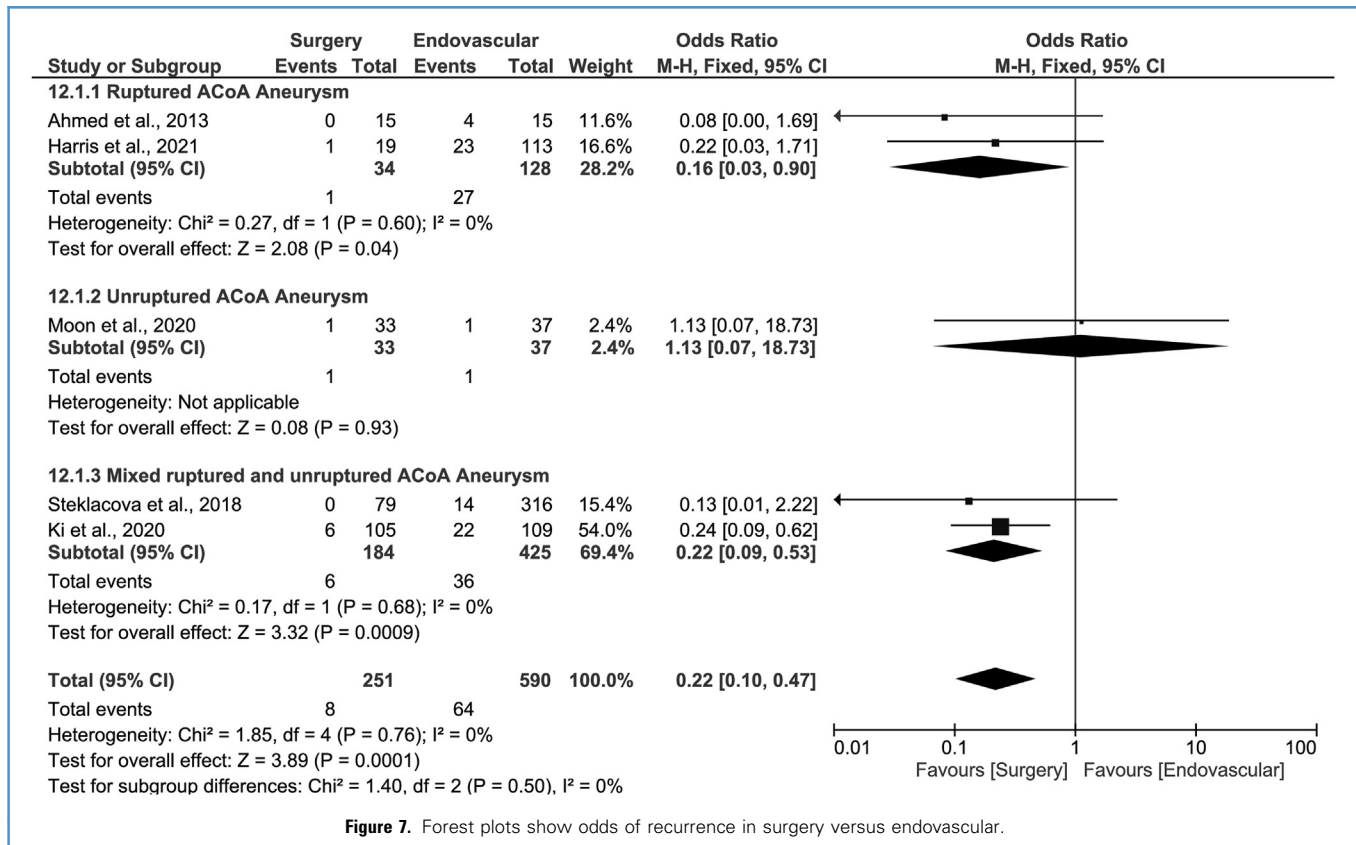
The results of this study show that surgical and endovascular management of ACoA aneurysm exhibit, in general, similar risk for postoperative morbidity and mortality. However, patients who underwent surgery for their ACoA aneurysm had higher odds of aneurysm obliteration and, conversely, lower odds of re-treatment and aneurysm recurrence (Table 3).

Morbidity and Mortality

The treatment goal for aneurysms is to remove the aneurysm from circulation without decreasing the baseline functional status of the patient. Although both surgical and endovascular approaches to aneurysms present risk to the patient,³ the results of this study indicate that the odds of post-treatment mortality and poor mRS (i.e., mRS > 2) are similar between the 2 treatment modalities. These findings align with the results of the Barrow Ruptured Aneurysm Trial

trial, which found similar risks of morbidity and mortality between surgical and endovascular management of aneurysms during a 10-year follow-up duration.⁹⁻¹² Furthermore, these findings are in line with studies by Koivisto et al. and Li et al., which too found comparable morbidity and survival albeit at a much shorter follow-up interval at 1 year.^{49,50} Although the International Subarachnoid Aneurysm Trial reported better outcomes following endovascular management of aneurysms, a statistically significant difference in outcomes was only identified along the internal carotid territory,^{7,8,17} suggesting the importance of stratifying outcome studies for aneurysms by the location of the aneurysm.

Many factors may predispose patients to poor outcomes during and after treatment for aneurysms. In particular, old age and the presence of subarachnoid hemorrhage (Hunt and Hess grade > II) have been shown to be associated with poor outcomes



following treatment for aneurysms.^{9,10,22,30,48} Additionally, the dome projection of the aneurysm too may impact treatment outcomes, with some

studies showing higher rates of procedural complications and mortality with superior, inferior, and posterior fundus projections. In these studies, the authors posited that

the position of the dome projection may interfere with the dissection pathway and thereby lead to intraoperative rupture or injury to adjacent vessels.^{30,51,52}

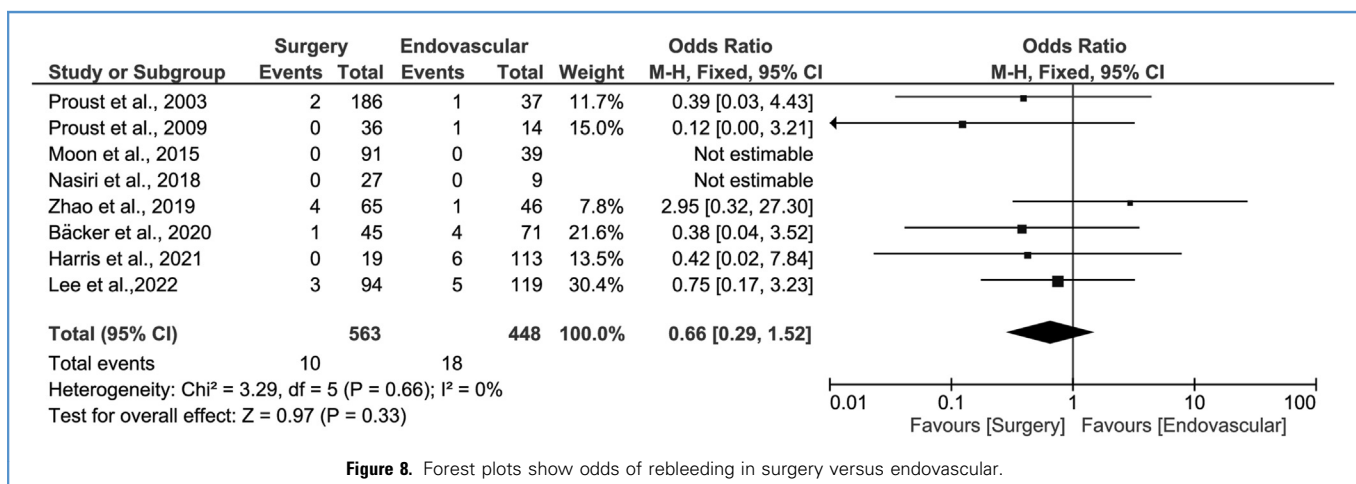


Table 3. Summary of Findings Table**Surgery versus Endovascular for Treatment of Ruptured ACoA Aneurysm**

Patient or Population: Ruptured ACoA Aneurysm

Setting: Inpatient

Intervention: Surgery

Comparison: Endovascular

Anticipated Absolute effects* (95% CI)						
Outcomes	Risk with Endovascular	Risk with Surgery	Relative effect (95% CI)	No of participants (Studies)	Certainty of the evidence (GRADE)	Comments
Mortality	120 per 1000	111 per 1000 (78–156)	OR 0.92 (0.62–1.36)	974 (10 observational studies)	⊕⊕⊕○ Moderate ^a	Lower score indicates lower odds of post-treatment mortality.
Follow-up mRS>2	396 per 1000	336 per 1000 (243–441)	OR 0.77 (0.49–1.20)	539 (6 observational studies)	⊕⊕⊕○ Moderate ^a	Lower score indicates lower odds of follow-up mRS>2.
Aneurysm Obliteration	786 per 1000	906 per 1000 (830–949)	OR 2.61 (1.33–5.10)	441 (4 observational studies)	⊕⊕⊕⊕ High ^{a,b}	Higher score indicates higher odds of aneurysm obliteration following treatment.
Retreatment	106 per 1000	36 per 1000 (13–96)	OR 0.31 (0.11–0.89)	409 (4 observational studies)	⊕⊕⊕○ Moderate ^{a,b,c}	Lower score indicates lower odds of retreatment following treatment.
Recurrence	211 per 1000	41 per 1000 (8–194)	OR 0.16 (0.03–0.90)	162 (2 observational studies)	⊕⊕⊕○ Moderate ^{a,b,c}	Lower score indicates lower odds of recurrence following treatment.

ACoA, anterior communicating artery; CI, confidence interval; OR, odds ratio.

GRADE Working Group grades of evidence.

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Explanations

a. Downgraded due to inclusion of non-randomized studies.

b. Upgraded due to large effect size.

c. Downgraded because the optimal information size (OIS) criterion was not met.

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

Obliteration, Re-Treatment, and Recurrence

Both endovascular and surgical management for aneurysms remove the aneurysm sac from circulation, thereby preventing rupture and/or rebleeding.^{3,30} The results of this meta-analysis show that the odds of aneurysm obliteration was significantly higher for patients who underwent surgery. Furthermore, the odds of re-treatment and aneurysm recurrence was lower for the surgical cohort. An endovascular approach to aneurysms may elevate the risk of residual aneurysms, which may lead to aneurysm recanalization, growth, and recurrent SAH. As the average rebleeding time for recurrent aneurysms is 8 years,¹⁰ patients undergoing endovascular management for aneurysms should therefore be vigilantly monitored for re-treatment. Considering that endovascular management of aneurysms is associated with a higher risk of residual aneurysms, young patients may be better managed microsurgically.

Limitations

We acknowledge several limitations in this study. Only articles with English text were selected from Medline, PubMed, and Embase. Furthermore, the included studies were nonrandomized and thereby present risk of selection bias. In addition, the total patient cohort of this systematic review included 23.5% unruptured ACoA aneurysm; however, a subgroup analysis was conducted on exclusively ruptured ACoA aneurysm to address this issue and graded the level of the evidence of meta-analysis outcomes for both the exclusively ruptured ACoA aneurysm and overall cohorts. Furthermore, when it was possible, the subgroup analyses per unruptured aneurysm have been conducted. Additionally, the included studies had different follow-up timelines; however, the outcomes incorporated into the meta-analysis was reported in range of 3–12 months. Furthermore, evaluating the long-term outcomes was not feasible. Moreover, comparing surgery versus endovascular treatment solely based on mRS as an index of the functional status is not enough. Other factors such as quality of life and the proportion of patients who return to their baseline functional status should be evaluated.

CONCLUSIONS

ACoA aneurysm can be safely treated both via surgical and endovascular management. Microsurgical clipping, however, may lead to higher rates of aneurysm obliteration and lower risk of recurrence and re-treatment. An intention-to-treat randomized controlled trial comparing surgery and endovascular treatment for ACoA aneurysm may assess which approach is most optimal for long-term outcomes.

REFERENCES

- Kassell NF, Torner JC, Haley ECJ, Jane JA, Adams HP, Kongable GL. The International Cooperative study on the timing of aneurysm surgery. Part 1: overall management results. *J Neurosurg.* 1990;73:18-36.
- Andaluz N, Van Loveren HR, Keller JT, Zuccarello M. Anatomic and clinical study of the orbitopterional approach to anterior communicating artery aneurysms. *Neurosurgery.* 2003;52:1140-1149.
- Brisman JL, Song JK, Newell DW. Cerebral aneurysms. *N Engl J Med.* 2006;355:928-939.
- Brilstra EH, Rinkel GJ, van der Graaf Y, van Rooij WJ, Algra A. Treatment of intracranial aneurysms by embolization with coils: a systematic review. *Stroke.* 1999;30:470-476.
- Byrne JV, Molyneux AJ, Brennan RP, Renowden SA. Embolisation of recently ruptured intracranial aneurysms. *J Neurol Neurosurg Psychiatry.* 1995;59:616-620.
- Nichols DA. Endovascular treatment of the acutely ruptured intracranial aneurysm. *J Neurosurg.* 1993;79:1-2.
- Molyneux A, Kerr R, Stratton I, et al. International Subarachnoid Aneurysm Trial (ISAT) of neurosurgical clipping versus endovascular coiling in 2143 patients with ruptured intracranial aneurysms: a randomised trial. *Lancet.* 2002;360:1267-1274.
- Molyneux AJ, Kerr RSC, Yu L-M, et al. International subarachnoid aneurysm trial (ISAT) of neurosurgical clipping versus endovascular coiling in 2143 patients with ruptured intracranial aneurysms: a randomised comparison of effects on survival, dependency, seizures, rebleeding, subgroups, and. *Lancet.* 2005;366:809-817.
- McDougall CG, Spetzler RF, Zabramski JM, et al. The Barrow ruptured aneurysm trial. *J Neurosurg.* 2012;116:135-144.
- Spetzler RF, McDougall CG, Albuquerque FC, et al. The Barrow ruptured aneurysm trial: 3-year results. *J Neurosurg.* 2013;119:146-157.
- Spetzler RF, McDougall CG, Zabramski JM, et al. The Barrow ruptured aneurysm trial: 6-year results. *J Neurosurg.* 2015;123:609-617.
- Spetzler RF, McDougall CG, Zabramski JM, et al. Ten-year analysis of saccular aneurysms in the Barrow ruptured aneurysm trial. *J Neurosurg.* 2019;132:771-776.
- Li H, Pan R, Wang H, et al. Clipping versus coiling for ruptured intracranial aneurysms: a systematic review and meta-analysis. *Stroke.* 2013;44:29-37.
- Fotakopoulos G, Tsianaka E, Fountas K, Makris D, Spyrou M, Hernesniemi J. Clipping versus coiling in anterior circulation ruptured intracranial aneurysms: a meta-analysis. *World Neurosurg.* 2017;104:482-488.
- Eide PK, Sorteberg A, Nome T, Rønning PA, Sorteberg W. Early surgical versus endovascular repair of ruptured blood-blister aneurysm of the internal carotid artery: a single-center 20-year experience. *J Neurosurg.* 2022;137:1766-1775.
- Raymond J, Kotowski M, Darsaut TE, Molyneux AJ, Kerr RS. Ruptured aneurysms and the International Subarachnoid Aneurysm Trial (ISAT): what is known and what remains to be questioned. *Neurochirurgie.* 2012;58:103-114.
- Molyneux AJ, Birks J, Clarke A, Sneade M, Kerr RSC. The durability of endovascular coiling versus neurosurgical clipping of ruptured cerebral aneurysms: 18 year follow-up of the UK cohort of the International Subarachnoid Aneurysm Trial (ISAT). *Lancet.* 2015;385:691-697.
- Mckissock W, Richardson A, Walsh L. "Posterior-communicating" aneurysms: a controlled trial of the conservative and surgical treatment of ruptured aneurysms of the internal carotid artery at or near the point of origin of the posterior communicating artery. *Lancet.* 1960;275:1203-1206.
- Mckissock W, Richardson A, Walsh L. Anterior communicating aneurysms: a trial of conservative and surgical treatment. *Lancet.* 1965;285:873-876.
- Mckissock W, Richardson A, Walsh L. Middle-cerebral aneurysms further results in the controlled trial of conservative and surgical treatment of ruptured intracranial aneurysms. *Lancet.* 1962;280:417-421.
- Kayembe KN, Sasahara M, Hazama F. Cerebral aneurysms and variations in the circle of Willis. *Stroke.* 1984;15:846-850.
- Nathal E, Yasui N, Sampei T, Suzuki A. Intra-operative anatomical studies in patients with aneurysms of the anterior communicating artery complex. *J Neurosurg.* 1992;76:629-634.
- Finitis S, Anxionnat R, Lebedinsky A, et al. Endovascular treatment of ACom intracranial aneurysms. Report on series of 280 patients. *Interu Neuroradiol.* 2010;16:7-16.
- Fang S, Brinjikji W, Murad MH, Kallmes DF, Cloft HJ, Lanzino G. Endovascular treatment of anterior communicating artery aneurysms: a systematic review and meta-analysis. *AJNR Am J Neuroradiol.* 2014;35:943-947.
- Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJWV, eds. *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3. Hoboken, NJ: John Wiley & Sons; 2022.

26. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *J Clin Epidemiol*. 2021; 134:178-189.
27. Farrell B, Godwin J, Richards S, Warlow C. The United Kingdom transient ischaemic attack (UK-TIA) aspirin trial: final results. *J Neurol Neurosurg Psychiatry*. 1991;54:1044-1054.
28. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. *BMJ*. 2016; 355:i4919.
29. Guyatt G, Oxman AD, Akl EA, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol*. 2011;64:383-394.
30. Proust F, Debono B, Hannequin D, et al. Treatment of anterior communicating artery aneurysms: complementary aspects of microsurgical and endovascular procedures. *J Neurosurg*. 2003;99: 3-14.
31. Proust F, Martinaud O, Gèrardin E, et al. Quality of life and brain damage after microsurgical clip occlusion or endovascular coiling versus surgical clipping in the treatment of ruptured anterior communicating artery aneurysms: Neuropsychological assessment - clinical article. *J Neurosurg*. 2009;110:19-29.
32. Ahmed AZ, Zohdi AM, Zaghloul MS, Elsamman AK. Endovascular coiling versus surgical clipping in the treatment of ruptured anterior communicating artery aneurysm in Cairo University Hospitals. *Egypt J Radiol Nucl Med*. 2013; 44(3):523-530.
33. Moon K, Levitt MR, Almefty RO, et al. Treatment of ruptured anterior communicating artery aneurysms: equipoise in the endovascular era? *Neurosurgery*. 2015;77:566-571 [discussion: 571].
34. Heit JJ, Ball RL, Telischak NA, et al. Patient outcomes and cerebral infarction after ruptured anterior communicating artery aneurysm treatment. *AJNR Am J Neuroradiol*. 2017;38:2119-2125.
35. Kumar P, Jain SK, Sinha VD, Srivastava T. Anterior communicating artery aneurysm clipping versus coiling: a comparative study of 50 cases. *Rom Neurosurg*. 2017;368-377.
36. Steklacova A, Bradac O, de Lacy P, Lacman J, Charvat F, Benes V. "Coil mainly" policy in management of intracranial ACoA aneurysms: single-centre experience with the systematic review of literature and meta-analysis. *Neurosurg Rev*. 2018; 41:825-839.
37. Nassiri F, Workewych AM, Badhiwala JH, Cusimano MD. Cognitive outcomes after anterior communicating artery aneurysm repair. *Can J Neurol Sci*. 2018;45:415-423.
38. Schmalz PGR, Enriquez-Marulanda A, Alturki A, Stapleton CJ, Thomas AJ, Ogilvy CS. Combined outcomes of endovascular or surgical treatment of unruptured anterior communicating artery aneurysms: is a more aggressive management strategy warranted? *World Neurosurg*. 2018;115:e331-e336.
39. Zhao B, Xing H, Fan L, et al. Endovascular coiling versus surgical clipping of very small ruptured anterior communicating artery aneurysms. *World Neurosurg*. 2019;126:e1246-e1250.
40. Bäcker HC, Shoap S, Vajda J, Nyáry I. Anterior communicating artery aneurysm rupture and functional outcome in short-term: clipping versus coiling. *J Integr Neurosci*. 2020;19:349-354.
41. Ki HJ, Lee KS, Kim BS, Choi JH, Shin YS. Clinical and morphological risk factors for the recurrence of anterior communicating artery aneurysms after clipping or coiling. *Acta Neurochir*. 2020;162: 2245-2250.
42. Moon JS, Choi CH, Lee TH, Ko JK. Result of coiling versus clipping of unruptured anterior communicating artery aneurysms treated by a hybrid vascular neurosurgeon. *J Cerebrovasc Endovasc Neurosurg*. 2020;22:225-236.
43. Baykara E, Topçu A. Comparing the frequency of symptomatic cerebral vasospasm and vasospasm-related ischemia in Fisher 3 grade ruptured anterior communicating artery aneurysms treated via microsurgical clipping or endovascular coiling. *Interdiscip Neurosurg Adv Tech Case Manag*. 2022;27, 101379.
44. Harris L, Hill CS, Elliot M, Fitzpatrick T, Ghosh A, Vindlacheruvu R. Comparison between outcomes of endovascular and surgical treatments of ruptured anterior communicating artery aneurysms. *Br J Neurosurg*. 2021;35:313-318.
45. Ma N, Feng X, Wu Z, Wang D, Liu A. Cognitive impairments and risk factors after ruptured anterior communicating artery aneurysm treatment in low-grade patients without severe complications: a multicenter retrospective study. *Front Neurol*. 2021;12:613785.
46. Haeren R, Hafez A, Korja M, Raj R, Niemelä M. Fast Transition from open surgery to endovascular treatment of unruptured anterior communicating artery aneurysms-A retrospective analysis of 128 patients. *World Neurosurg*. 2022;165:e668-e679.
47. Lee SH, Park JS. Outcome of ruptured anterior communicating artery aneurysm treatment compared between surgical clipping and endovascular coiling: a single-center analysis. *Medicine (Baltimore)*. 2022;101:e30754.
48. Ogawa A, Suzuki M, Sakurai Y, Yoshimoto T. Vascular anomalies associated with aneurysms of the anterior communicating artery: microsurgical observations. *J Neurosurg*. 1990;72:706-709.
49. Koivisto T, Vanninen R, Hurskainen H, Saari T, Hernesniemi J, Vapalahti M. Outcomes of early endovascular versus surgical treatment of ruptured cerebral aneurysms. A prospective randomized study. *Stroke*. 2000;31:2369-2377.
50. Li Z-Q, Wang Q-H, Chen G, Quan Z. Outcomes of endovascular coiling versus surgical clipping in the treatment of ruptured intracranial aneurysms. *J Int Med Res*. 2012;40:2145-2151.
51. Perlmutter D, Rhoton ALJ. Microsurgical anatomy of the anterior cerebral-anterior communicating-recurrent artery complex. *J Neurosurg*. 1976;45: 259-272.
52. Vincentelli F, Lehman G, Caruso G, Grisoli F, Rabehanta P, Gouaze A. Extracerebral course of the perforating branches of the anterior communicating artery: microsurgical anatomical study. *Surg Neurol*. 1991;35:98-104.

Presentation: This paper was presented in 91st American Association of Neurological Surgeons (AANS) annual meeting on April, 2023, in Los Angeles, CA, USA.

Conflict of interest statement: The authors declare that the article content was composed in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Received 6 March 2023; *accepted* 27 March 2023

Citation: *World Neurosurg*. (2023) 175:31-44.
<https://doi.org/10.1016/j.wneu.2023.03.111>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

1878-8750/\$ - see front matter Published by Elsevier Inc.